

International AS and A-level

Chemistry (9620)

CH04

Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH04 - JANUARY 2025

Students were more confident with the mathematical questions in this paper than questions requiring a detailed explanation. Students were also generally very good at giving reagents and conditions needed for the reactions covered. Students were less good at drawing mechanisms than they have been in previous series with many students drawing incorrect curly arrows and missing lone pairs on nucleophiles.

QUESTION 01

- 01.1 A number of students wrote the reduction equation with only one [H] and/or with the incorrect structural formula for pentan-2-one. Some gave butanone and others pentan-3-one. A molecular formula for the organic reactant or product was not sufficient for the second mark. When using LiAlH_4 students needed to state that it would be in ether to gain the mark.
- 01.2 A number of students drew the curly arrow from the C of the $\text{C}=\text{O}$ rather than from the double bond. A few students drew the curly arrow from the N atom of the cyanide ion rather than the lone pair on the C atom. Most students were able to name the mechanism correctly.
- 01.3 Some gave the substituents in the wrong order (eg 2-methyl-2-hydroxy) rather than correctly in alphabetical order. Some gave the name as 2-hydroxy-2-methylpentannitrile rather than 2-hydroxy-2-methylpentanenitrile. A few answers suggested that the students had not included the C of the CN and gave the name 2-hydroxy-2-methylbutanenitrile.
- 01.4 Some students did not state that the enantiomers would be in equal amounts for M1. There were a number of students who stated that the $\text{C}=\text{O}$ bond was planar rather than the carbonyl group and a few stated that the molecule was planar rather than just the carbonyl group. For M3 not all students said that attack was equally likely from either side of the carbonyl group.
- 01.5 Most students were able to identify an isomer that did not have a chiral carbon.
- 01.6 Most students gave the appropriate test for distinguishing aldehydes and ketones. Some did not realise that acidified potassium dichromate could not be used in this case as there was also an -OH group present in both compounds. Responses were credited where students noted that the question involved reducing sugars and that these would give a positive result with Benedict's solution.

QUESTION 02

- 02.1 Most students were able to write this equation correctly.
- 02.2 Only around 20% students scored both marks for this question. Some students did not explain what increasing the concentration meant (M1) and some incorrectly said that there were more molecules or atoms per unit area which was a chemical error as the question was about increasing the concentration of iodide ions. Some students stated that the frequency of collisions increased but didn't mention successful collisions.

02.3 Over 50% students gained all four marks for this question. Those who did not missed marks for not converting the temperature and/or the activation energy. A few were not able to rearrange the equation and seemed unfamiliar with how to evaluate $e^{\frac{-E_a}{RT}}$.

02.4 Most students were able to correctly deduce the rate equation. Some did not show how they used the data to determine the order of reaction so were not credited with full marks.

02.5 Most students recognised that the graph showed a zero-order reaction, but some stated it was first order because it was a straight line.

02.6 Over 80% students answered this question correctly, but a few multiplied the uncertainty by two which is unnecessary for a single reading.

QUESTION 03

03.1 Only around one-third of students gained full marks on this question. Many failed to mention that it was the rate of reaction that increased, and others did not state that the rate of the forward and backward reaction were increased equally.

03.2 Most students were awarded the mark for this question.

03.3 Around 90% students got this correct.

03.4 Over two-thirds of students got both marks for this question. The main error seen was using square brackets (i.e. concentrations of the reactants and products) rather than partial pressures.

03.5 Most students gained marks on this item but fewer than for 03.3 suggesting that students are less familiar with the effect of changing conditions on the value of K_p compared to the effect on position of equilibrium.

03.6 Over 70% of students received all four marks for this question.

QUESTION 04

04.1 Around a third of students got all three marks for this question. A number of students did not include the numerical data in their answers, and some did not state that benzene was more stable because of the delocalisation of electrons.

04.2 Only around 20% students achieved this mark. A common error was to state that benzene would not react with bromine. Students needed to state that it doesn't react via electrophilic addition as benzene does react with bromine with free-radical addition.

04.3 Most students who did not achieve full marks on this question got the mechanism incorrect with a number not stating it was a free-radical reaction. Some incorrectly showed the delocalised ring structure remaining.

04.4 Around three-quarters of students answered this correctly.

QUESTION 05

05.1 Around 40% students did not state that the acids both needed to be concentrated.

05.2 Most students were able to identify the reaction as a reduction and give a correct reducing agent.

05.3 Over 80% got this question correct showing confidence in carbon-13 NMR.

05.4 Many students did not state that the aromatic amines are used in manufacture of dyes and simply wrote dyes.

05.5 Under 30% of students achieved all three marks in this question. Some of the mechanisms drawn did not show clearly enough where pairs of electrons were moving from and moving to. A number of students did not show the lone pair on the N atom of methylamine. A few drew the second arrow from C to the Cl atom rather than from the C-Cl bond. A surprising number of students drew the arrow on the intermediate going from the N-H bond to the H rather than to the positively charged N.

05.6 Over half of students got both marks for this question. The most common error was stating that the mechanism was nucleophilic substitution rather than addition-elimination.

QUESTION 06

06.1 Around 80% gave the correct zwitterion structure.

06.2 and 6.3 Most students got these questions correct.

06.4 Around 60% students got this correct. A number of students did not show the correct bonding between atoms so could not be awarded the mark.

06.5 Around three-quarters of students got both marks for this question. A number of students did not identify that it was hydrogen bonding holding the structure together.

06.6 Just under 10% of students did not attempt this question. Of those that did answer, most stated that the lysine had a greater affinity for the stationary phase but did not explain why lysine is more polar than glycine.

QUESTION 07

07.1 Most students achieved both marks for this question.

07.2 Few students got all three marks for this question. Common errors were not referring to the bonds in the chain of the polymer and not being specific about the bonds that would break if the polymer were to breakdown: C-O in terylene and C-C in poly(propene).

QUESTION 08

08.1 A number of students stated the condition used was heat and were not specific about reflux being needed. Some students did not state that the potassium dichromate needed to be acidified.

08.2 Less than 30% of students got all four marks for this question. Most were able to identify HCl as the other product, but many were not able to draw the mechanism correctly. As with the previous mechanism on the paper a number did not draw curly arrows to show movement of electrons from either a lone pair or from a bond.

08.3 Most were able to name the ester. The most common error was to miss the carbonyl carbon and name it ethyl ethanoate.

08.4 Most students got this question correct.

08.5 A number of students incorrectly stated that the propanoic acid didn't have a C-O stretch and so did not get both marks for this question. Around 20% of students gained both marks.

08.6 Just over half of students scored both marks for this question. The most common error was not using the correct terminology for the splitting pattern. Students need to know the key terms singlet, doublet, triplet and quartet.

08.7 Just under half of students drew both structures needed for this question. More students were able to identify the carboxylic acid than the ester.